

Zurich ServiceNow AI Platform Capabilities

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Service Graph Connector for Microsoft Azure

Use the Service Graph Connector for Microsoft Azure to pull data from Microsoft Azure into your CMDB.

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

Supported versions

Supported ServiceNow versions:

- Xanadu
- Yokohama
- Zurich

Use cases

The following are examples on how you can use the Service Graph Connector:

- Visibility into cloud resources, relationships, and state in near real-time.
- Service ITAM/SAM outcomes through deep discovery of applications.

Configuring a connection for the connector

You can configure a connection for the connector by using the SGC Central view in the CMDB Workspace. The view enables you to discover and install connectors, and then effectively manage the full life cycle of creating, editing, monitoring, and debugging connections. To configure the connector using SGC Central, see [Configure Service Graph Connector for Microsoft Azure using SGC Central](#).

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

CMDB integrations dashboard

The Integration Commons for CMDB store app provides a dashboard with a central view of the status, processing results, and processing errors of all installed integrations. You can see metrics for all integration runs. You can filter the view to a specific CMDB integration, a specific time duration, or a specific integration run. For more details about monitoring Microsoft Azure integrations in the CMDB Integrations Dashboard, see [Using the CMDB Integrations Dashboard](#).

Data mapping

Data from the Azure data sources is mapped and transformed into the ServiceNow CMDB Configuration Item (CI) class definitions using the Robust Transform Engine (RTE). Data is inserted into the ServiceNow® CMDB using the Identification and Reconciliation Engine (IRE).

When you complete setting up the connection, you can configure the integration to periodically pull data from Azure. The data is loaded into staging tables and then inserted into the target tables.

The following table lists the data sources, the staging tables, and the target tables as CMDB CI and non-CMDB classes for Azure.

Data mapping for Azure

Data source	Staging table	Target tables
SG-Azure Subscriptions	SG Azure Subscriptions [sn_sg_azure_integ_subscriptions]	Cloud Service Account Key Value
SG-Azure Datacenter Updation	SG-Azure Datacenter Updation [sn_sg_azure_integ_datacenter_updation]	Azure Datacenter
SG-Azure Resource Group	SG-Azure Resource Group [sn_sg_azure_integ_resource_groups]	Resource Group

Data source	Staging table	Target tables
		Key Value
SG-Azure Availability Zone	SG-Azure Availability Sets [sn_sg_azure_integ_availability_sets]	Availability Zone Key Value
SG-Azure Storage Volume	SG-Azure Storage Volume [sn_sg_azure_integ_storage_volume]	Storage Volume Key Value
SG-Azure Load Balancers	SG-Azure Load Balancers [sn_sg_azure_integ_load_balancers]	Cloud Load Balancer Cloud LB IPAddress Key Value
SG-Azure Network	SG-Azure Network [sn_sg_azure_integ_network]	Cloud Network Cloud Service Account Cloud Subnet Key Value
SG-Azure Public IP Address	SG-Azure Public IP Address [sn_sg_azure_integ_public_ip_address]	Cloud Public IP Address Cloud Mgmt Network Interface Cloud Load Balancer

Data source	Staging table	Target tables
		Cloud LB IPAddress Key Value
SG-Azure Security Group	SG-Azure Security Group [sn_sg_azure_integ_security_group]	Compute Security Group Key Value
SG-Azure Storage Accounts	SG-Azure Storage Accounts [sn_sg_azure_integ_storage_accounts]	Cloud Storage Account Key Value
SG-Azure Virtual Machines	SG-Azure Virtual Machine [sn_sg_azure_integ_virtual_machine]	Virtual Machine Instance Image Computer Linux Server Server Windows Server Key Value
SG-Azure Network Interface	SG-Azure Network Interface [sn_sg_azure_integ_network_interface]	Cloud Mgmt Network Interface IP Address

Data source	Staging table	Target tables
		Virtual Machine Instance Key Value
SG-Azure SQL	SG-Azure SQL [sn_sg_azure_integ_sql]	Cloud DataBase Key Value
SG-Azure Functions	SG-Azure Functions [sn_sg_azure_integ_functions]	Cloud Function
SG-Azure Generic Resources	SG-Azure Generic Resources [sn_sg_azure_integ_generic_resources] Note: The SG-Azure Generic Resources data source imports data for generic resources that aren't tracked by other data sources.	Cloud Resource Key Value
SG-Azure Scale Sets	SG-Azure Scale Sets [sn_sg_azure_integ_scale_sets]	Instance Scale Set
SG-Azure Scale Sets VMs	SG-Azure Scale Sets VMs [sn_sg_azure_integ_scale_sets_vms]	Virtual Machine Instance Storage Volume

Data source	Staging table	Target tables
		Image Cloud Mgmt Network Interface Hardware Type Azure Datacenter Linux Server Server Windows Server Key Value
SG-Azure Software	SG-Azure Software [sn_sg_azure_integ_software] Note: The SG-Azure Software data source imports the software data from the ConfigurationDat a table in the Log Analytics workspace of Azure. The SG-Azure Software import is set to inactive by default.	When the Software Asset Management (SAM) application isn't installed: Software Software Instance When the SAM application is installed: Software Installation

Data source	Staging table	Target tables
SG-Azure Software Remove	SG-Azure Software Remove [sn_sg_azure_integ_software_remove] The SG-Azure Software Remove import is set to inactive by default.	When the SAM application isn't installed: Software Software Instance When the SAM application is installed: Software Installation
SG-Azure TCP	SG-Azure TCP [sn_sg_azure_integ_tcp]	TCP Connection Running Process

Data source	Staging table	Target tables
	Note: The SG-Azure TCP data source imports the TCP and processes data from the VMConnection and VMProcess tables in the Log Analytics workspace of Azure. To populate the tables, you must first enable the VM insights feature in the Azure portal. For more information, see Enabling VM insights for collecting Azure data .	
SG-Azure Hardware Template Updation	Hardware templates [sn_sg_azure_integ_hardware_templates]	Hardware Type
SG-Azure HW Consolidation	SG-Azure HW Consolidation [sn_sg_azure_integ_hw_consolidation]	Virtual Machine Instance Computer
SG-Azure Run Command	SG-Azure Run Command [sn_sg_azure_integ_run_command]	No data created

Data source	Staging table	Target tables
SG-Azure Get Run Command	SG-Azure Get Run Command [sn_sg_azure_integ_get_run_command]	Application TCP Connection Running Process
SG-Azure Kubernetes Cluster	SG-Azure Kubernetes Cluster [sn_sg_azure_integ_kubernetes_cluster]	Kubernetes Cluster

- The connector utilizes the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table to differentiate between generic and other supported resource types.

The connector first populates all supported resources in the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table. These resources, categorized under their respective supported resource types, have specific data sources designated for ingestion within the connector. When the SG-Azure Generic Resources data source is executed and retrieves unsupported resource types, they are added to the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table and categorized as generic.

- Starting with the Service Graph Connector for Microsoft Azure 1.12.0 version, the SG-Azure Hardware Template Updation and SG-Azure HW Consolidation import schedules replace the SG-Azure Hardware Template import schedule. The SG-Azure Hardware Template Updation and SG-Azure HW Consolidation data sources fetch only the data of the hardware templates that are used for VMs in Azure, instead of fetching all hardware templates.
- Starting with the Service Graph Connector for Microsoft Azure 1.14.0 version, the SG-Azure SQL data source populates the microsoft.dbformysql/flexibleservers and microsoft.dbforpostgresql/flexibleservers resource types in the Cloud Database

[cmdb_ci_cloud_database] table instead of the Cloud Resource [cmdb_ci_cmp_resource] table.

After upgrading to version 1.14.0, run a full data load to migrate all active flexible servers from the Cloud Resource [cmdb_ci_cmp_resource] table to the Cloud Database [cmdb_ci_cloud_database] table.

- Starting with the Service Graph Connector for Microsoft Azure 1.14.0 version, all data sources except SG-Azure Scale Sets VMs are migrated from Azure Resource Manager (ARM) APIs to Resource Graph APIs for a full data load. Microsoft Azure doesn't support uniform orchestration-type Scale Set VMs on Resource Graph APIs.

For more information on where data is saved when pulling data from Azure, see [CMDDB classes targeted in Service Graph Connector for Microsoft Azure](#) and [Supported Microsoft Azure resource types](#).

You can use the IntegrationHub ETL app to view the data maps. See [IntegrationHub ETL](#) for more information.

Event-based discovery

The Service Graph Connector for Microsoft Azure works with event-based discovery. Events that are pulled from Azure can create and update CIs that were brought by the Service Graph Connector for Microsoft Azure.

Additional resource

[How do I configure the Azure Service Graph Connector?](#)

Related tasks

- [Configure Service Graph Connector for Microsoft Azure using the guided setup](#)

Related concepts

- [Service Graph Connectors](#)
- [Configuring data synchronization in Service Graph Connector for Microsoft Azure](#)

- [Enabling VM insights for collecting Azure data](#)
- [Setting up multiple connections in Service Graph Connector for Microsoft Azure](#)

Related reference

- [CMDB classes targeted in Service Graph Connector for Microsoft Azure](#)

Configure Service Graph Connector for Microsoft Azure using SGC Central

Use the playbook available with the SGC Central application to set up the Service Graph Connector for Microsoft Azure for pulling in Microsoft Azure data into the CMDB.

Before you begin

Install Service Graph Connector for Microsoft Azure version 1.9.0 or later from the ServiceNow Store. For ServiceNow Store installation steps, see [Install a ServiceNow Store application](#).

Important: After the initial import of Azure data, the application performs a delta import of data. However, you can perform a full pull of data anytime later by disabling the delta synchronization and setting the `sn_sg_azure_integ.is_delta_sync_enabled` property to `false`. For more information, see [Configuring data synchronization in Service Graph Connector for Microsoft Azure](#).

Role required: The following table shows the roles required for each stage of the playbook.

Role required for each playbook stage

Stage	Role
Prerequisites	admin
Setup	cmdb_inst_admin or admin

About this task

The playbook experience for onboarding connectors is activated with SGC Central in the CMDB Workspace. To configure the SGC Central application, see [Configuring SGC Central](#) and for more information on how to interact with a playbook, see [Interact with Playbook](#).

The connector uses the Azure Management APIs for the complete pull of data from Azure. However, to pull delta changes from Azure, the Azure Resource Graph APIs are used. The domain name system (DNS) is Microsoft Azure Management, but the path is a resource graph.

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. In the CMDB Workspace, select **SGC Central**.
3. On the Overview page, select **Create connection**.

Tip: Alternatively, you can select **Create connection** on the All connections page.
4. On the Create connection window, select the Microsoft Azure connector type and then select **Create connection**.
5. Complete the initial prerequisites when setting up a connection for the first time using a connector.

Note: This step is required only during the first-time setup. See [Perform initial setup tasks when creating a connection in SGC Central](#).
6. Complete the prerequisites for setting up the Azure environment.
 - a. Complete the instructions required for setting up the Azure environment.
 - a. In the **Prerequisites** stage of the playbook, select the **Review setup instructions** activity.
 - b. Obtain the OAuth credentials including client ID, client secret, and token URL from your Microsoft Azure administrator.

- c. For importing hardware data, grant the **User . Read** permission on the Microsoft Graph API in the Azure environment. Also, ensure that the IAM policy for the Azure subscription has the Reader role. See [Assign a user as an administrator of an Azure subscription with conditions](#) on the Microsoft Azure documentation site.
 - d. For importing software data, grant the **Data . Read** permission on the Log Analytics API in the Azure environment. In addition, set up a Log Analytics workspace in the Azure environment.
 - e. After completing the **Review setup instructions** activity, select **Continue**.
7. Determine whether to import only hardware data or both hardware and software data.
 - a. In the **Setup** stage of the playbook, select the **Select services** activity.
If needed, expand the **Setup** stage to select an activity.
 - b. (Optional) To import both hardware and software data, select the **Hardware and software** check box.

Note: Importing software data requires a hardware connection.

If you need to import only hardware data, the **Hardware only** check box is already selected by default.
- c. Select **Continue**.
8. Enter the connection details and test the API connection for importing hardware data.
 - a. In the **Setup** stage of the playbook, select the **Create and test hardware connection** activity.
 - b. On the form, fill in the fields.

Create and test hardware connection form

Field	Description
Connection name	Name to identify the hardware connection record. For example, SG-Azure hardware connection.
Connection URL	Base URL to connect to your Azure client application. Note: This field is automatically set to the URL to connect to the application. Leave the field value as is.
OAuth client ID	Application (client) ID of your Azure client application as noted in step 6.a.
OAuth client secret	Client secret of your Azure client application as noted in step 6.a.
OAuth token URL	Callback URL for the Azure client application. Based on the region of your Azure client application, enter the token URL in one of the following formats: - Global <pre>https://login.microsoftonline.com/<tenantid>/oauth2/v2.0/token</pre> - US Government

Field	Description
	https://login.microsoftonline.us/<tenantid>/oauth2/v2.0/token - China https://login.partner.microsoftonline.cn/<tenantid>/oauth2/v2.0/token - Germany https://login.microsoftonline.de/<tenantid>/oauth2/v2.0/token Where <tenantid> is the tenant ID of your Azure client application as noted in step 6.a.

- c. Select **Create and test connection**.
 - d. Once the connection test is complete, select **Continue**.
9. Configure the import schedule to import hardware data at regular intervals.
- a. In the **Setup** stage of the playbook, select the **Configure hardware import schedule** activity.
 - b. Select **Configure import schedule**.
 - c. Expand the Parent scheduled data import within the Import schedules list to select the **Connection name-SG-Azure Subscriptions** import schedule.
 - d. In the Configure import schedule dialog box, select the **Active** check box, and then fill in the run schedule and time details.

For more information, see [Schedule a data import](#).

- e. Select **Save**.
Alternatively, select **Execute Now** to execute the import schedule immediately.

- f. Select **Continue**.

10. Enter the connection details and test the API connection for importing software data.

The **Create and test software connection** activity for importing software data appears only when the **Hardware and software** check box is selected in step [7.b](#).

- a. In the **Setup** stage of the playbook, select the **Create and test software connection** activity.
- b. On the form, fill in the fields.

Create and test software connection form

Field	Description
Software connection name	Name to identify the software connection record. For example, SG-Azure software connection.
Hardware connection name	Name of the hardware connection record. Note: This field is automatically set to the hardware connection name specified in step 8.b .
Connection URL	Base URL to connect to the Log Analytics workspace in the following format:

Field	Description
	https://api.loganalytics.io/v1/workspaces/<workspace_id> Where <workspace_id> is the ID of the Log Analytics workspace as noted in step 6.a. Note: This field is automatically set to the URL to connect to the Log Analytics workspace. Replace the <workspace_id> variable in the auto-generated URL with the workspace ID of your Log Analytics workspace.
OAuth client ID	Application (client) ID of your Azure client application as noted in step 6.a.
OAuth client secret	Client secret of your Azure client application as noted in step 6.a.
OAuth token URL	Token URL of your Azure client application. Based on the region of your Azure client application, enter the token URL in one of the following formats: Global https://login.microsoftonline.com/<tenantid>/oauth2/v2.0/token

Field	Description
	- US Government https://login.microsoftonline.us/<tenantid>/oauth2/v2.0/token - China https://login.partner.microsoftonline.cn/<tenantid>/oauth2/v2.0/token - Germany https://login.microsoftonline.de/<tenantid>/oauth2/v2.0/token Where <tenantid> is the tenant ID of your Azure client application as noted in step 6.a.

- c. Select **Create and test connection**.
 - d. Once the connection test is complete, select **Continue**.
11. Configure the import schedule to import software data at regular intervals.

The **Configure software import schedule** activity for importing software data appears only when the **Hardware and software** check box is selected in step [7.b](#).

- a. In the **Setup** stage of the playbook, select the **Configure software import schedule** activity.
- b. Expand the Parent scheduled data import within the Import schedules list to select the **Connection name-SG-Azure TCP** import schedule.

- c. In the Configure import schedule dialog box, select the **Active** check box, and then fill in the run schedule and time details.

For more information, see [Schedule a data import](#).

- d. Select **Save**.
Alternatively, select **Execute Now** to execute the import schedule immediately.
- e. Repeat steps 11.b to 11.d for any other parent import schedule if not already set to active.
- f. Select **Continue**.

12. In the **Setup** stage of the playbook, select the **Connection setup complete** activity to verify whether the connection was created.

What to do next

Select **View all connections** to review the connection details. The created connection appears in the Installed connections list.

Related concepts

- [Service Graph Connector for Microsoft Azure](#)
- [Accessing the connection details of Service Graph Connector for Microsoft Azure](#)
- [Additional features within the Service Graph Connector for Microsoft Azure](#)

Related reference

- [CMDB classes targeted in Service Graph Connector for Microsoft Azure](#)
- [Supported Microsoft Azure resource types](#)
- [Service Graph Connector for Microsoft Azure properties](#)

Configure Service Graph Connector for Microsoft Azure using the guided setup

Set up data sources and scheduled import jobs to pull in data from Azure into your CMDB.

Before you begin

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

To use this Service Graph Connector, you need a subscription to a Subscription Unit that is based in the IT Operations Management (ITOM) Visibility application or in the ITOM Discovery application. As defined in the section titled "Managed IT Resource Types" in [ServiceNow Subscription Unit Overview](#) for your subscription, for managed IT resources that are created or modified in the CMDB by this Service Graph Connector, but that aren't yet managed by [ITOM Visibility](#) or [ITOM Discovery](#), these resources will increase Subscription Unit consumption from that application. Review your current Subscription Unit consumption within ITOM Visibility or ITOM Discovery to ensure available capacity.

Dependencies and requirements:

- The [Integration Commons for CMDB](#) store app, which is automatically installed.
- The CMDB CI class models store app, which is automatically installed. See [CMDB CI Class Models](#).
- Discovery Core plugin (com.snc.discovery.core), which is automatically installed by Discovery.
- The Datastream Action plugin (com.glide.hub.action_type.datastream), which is automatically installed.
- The ITOM Discovery License plugin (com.snc.itom.discovery.license). You must activate this plugin.

- ITOM Licensing plugin (com.snc.itom.license). For more information, see [Request Discovery](#).

Important: After the initial import of Azure data, the application performs a delta import of data. However, you can perform a full pull of data anytime later by disabling the delta synchronization and setting the `sn_sg_azure_integ.is_delta_sync_enabled` property to `false`. For more information, see [Configuring data synchronization in Service Graph Connector for Microsoft Azure](#).

Role required: admin

About this task

The connector uses the Azure Management APIs for the complete pull of data from Azure. However, to pull delta changes from Azure, the Azure Resource Graph APIs are used. The domain name system (DNS) is Microsoft Azure Management, but the path is a resource graph.

For more information on the Azure setup instructions, see the [Service Graph Connector for Azure - Overview](#) article on the ServiceNow Community site.

Procedure

1. Navigate to **All > Service Graph Connectors > Azure > Setup**.
2. On the Getting started page, select **Get Started**.
3. Create data sources and scheduled imports for the new connection.
 - a. On the Service Graph Connector for Microsoft Azure page, in the Update Data Sources and Scheduled Imports Access, select the task **Update Scheduled Data Import Access**.
 - b. In the Update Scheduled Data Import Access section, select **Configure** and do the following:
 - a. For the Update Scheduled Data Import access task, select **Configure**.
 - b. Select the **Global** application scope by using the application picker.

- c. In the Application Access related list, select the **Can create**, **Can update**, and **Can delete** check boxes, if not already selected.
 - d. Select **Update** to close the tab and return to the guided setup.
 - e. Modify the application scope again to **Service Graph Connector for Microsoft Azure** by using the application picker.
 - f. Set the Update Scheduled Data Import access task to complete by selecting **Mark as Complete**.
 - g. Repeat the steps 3.b.ii to 3.b.v for the Update Data Source Access task with the Data Source [sys_data_source] table and set the task to complete by selecting **Mark as Complete**.
- c. Clear the cache for the new connection.
 - a. In the Clear Cache for Datasource and Import set section, select **Configure**.
 - b. In the **Run script** text box of the background script page, enter the following script:

```
GlideTableManager.invalidateTable("sys_data_source");
    GlideCacheManager.flushTable("sys_data_source");

    GlideTableManager.invalidateTable("scheduled_import_set");
    GlideCacheManager.flushTable("scheduled_import_set");

    GlideTableManager.invalidateTable("sys_db_object");
    GlideCacheManager.flushTable("sys_db_object");
```

- c. Select **Run Script** to run the background script in the **global** scope.

The script may take several minutes to execute.

- d. After the script is executed, select **Close**.
 - e. From the application picker, select the **Service Graph Connector for Microsoft Azure** application.
 - f. Select **Mark as Complete**.
4. Create a connection to import hardware configuration items (CIs) from the Azure client application.

Note: Ensure that you have the `User.Read` permission on the Microsoft Graph API for the hardware import. Also, ensure that the IAM policy for the Azure subscription has the Reader role. See [Assign a user as an administrator of an Azure subscription with conditions](#) on the Microsoft Azure documentation site.

- a. Obtain the OAuth credentials from your Azure administrator. Make a note of the following details:

- Application (client) ID
- Client Secret
- Directory (tenant) ID
- Connection URL

Note: After getting the OAuth credentials, in the guided setup for Service Graph Connector for Microsoft Azure, go to the Create connection for the hardware import section of the Service Graph Connector for Microsoft Azure page and set the Get the OAuth credentials task to complete by selecting **Mark as Complete**.

- b. Configure your Azure hardware connection and credentials.
 - a. In the Create connection for the hardware import section of the Service Graph Connector for Microsoft Azure page, select **Continue**.
 - b. For the Create or Edit connection task, select **Configure**.
 - c. On the Connections page of the Workflow Studio, select **Configure** for the **SG-Azure Hardware Connection**

connection that is available by default for the hardware import.

You can create multiple connections by selecting **Add Connection**.

- d. On the form, review and modify the fields.

Configure Connection form

Field	Description
Connection Information	
Connection Name	Name to uniquely identify the hardware connection record. For example, SG-Azure Hardware Connection.
Connection URL	Base URL to connect to your Azure client application. Note: This field is automatically set to the URL to connect to the application. Leave the field value as is.
Credential Information	
OAuth Client ID	Application (client) ID of your Azure client application as described in step 4.a.
OAuth Client Secret	Client Secret of your Azure client application as described in step 4.a.
OAuth Token URL	Token URL of your Azure client application. Based on the region of your Azure client application,

Field	Description
	enter the token URL in one of the following formats: - Global https://login.microsoftonline.com/<tenantid>/oauth2/v2.0/token - US Government https://login.microsoftonline.us/<tenantid>/oauth2/v2.0/token - China https://login.partner.microsoftonline.cn/<tenantid>/oauth2/v2.0/token - Germany https://login.microsoftonline.de/<tenantid>/oauth2/v2.0/token Where <tenantid> is the tenant ID of your Azure client application as described in step 4.a.

e. Select **Configure and Get OAuth Token**.

f. When the OAuth token flow is successfully completed, return to the Create connection for the hardware import task page using the back button for your browser.

- g. Set the Create or Edit connection task to complete by selecting **Mark as Complete**.
- c. Test the Microsoft Graph API connection to import hardware data from the Azure client application.
 - a. For the Test Connection task, select **Configure**.
 - b. Select the data source associated with the newly created connection in the **Name** column of the Data Sources list.
 - c. Select the **Test Load 20 Records** related link.
 - d. When the state changes to **Complete**, return to the setup by selecting **Back to Guided Setup** in the Help panel.
 - e. Set the Test Connection task to complete by selecting **Mark as Complete**.
- d. Review the scheduled data imports configuration.
 - a. For the Set up scheduled import jobs task, select **Configure**.
 - b. Select the SG-Azure Subscriptions scheduled job.
 - c. On the Scheduled Data Import form, verify the field values for the scheduled job and select the **Active** check box.

For more information, see [Schedule a data import](#).
 - d. Select **Update**.
 - e. Set the Set up scheduled import jobs task to complete by selecting **Mark as Complete** in the Help panel.
5. Create a connection to import software information from the Azure client application.

Note: Ensure that you have the **Data . Read** permission on the Log Analytics API for the software import.

 - a. Obtain the OAuth credentials and set up the Log Analytics workspace in the Azure.
 - a. Ensure that you have the OAuth credentials from step [4.a](#).

- b. Configure the Log Analytics workspace in the Azure client application.

- Use an existing workspace, if available.
- Create another workspace.

For more information, see [Create Log Analytics workspace](#) in the Azure documentation.

Note: Make a note of the Workspace ID.

- c. Deploy Azure Monitor Agent (AMA), and then enable change tracking and inventory using AMA.

For more information, see [Enable Change Tracking and Inventory using Azure Monitoring Agent](#) in the Azure documentation and the Configuring Azure for monitoring your Azure VMs section of the [How do I configure the Azure Service Graph Connector? \[KB2226163\]](#) article in the Now Support Knowledge Base.

Note: After obtaining the OAuth credentials, in the guided setup for Service Graph Connector for Microsoft Azure, go to the Create connection for the software import section of the Service Graph Connector for Microsoft Azure page and set the Get the OAuth credentials task to complete by selecting **Mark as Complete**.

- b. Configure your Azure software connection and credentials.
 - a. In the Create connection for the software import section of the Service Graph Connector for Microsoft Azure page, select **Continue**.
 - b. For the Create or Edit connection task, select **Configure**.
 - c. On the Connections page of the Workflow Studio, select **Configure** for the **SG-Azure log analytics connection** connection that is available by default for the software import.

You can create multiple connections by selecting **Add Connection**.

- d. On the form, review and modify the fields.

Configure Connection form

Field	Description
Connection Information	
Software Connection Name	Name to uniquely identify the software connection record. For example, SG-Azure log analytics connection.
Hardware Connection Name	Name of the hardware connection associated with the software as described in step 4.b.iv. You add a hardware connection name to associate the software connection with the respective hardware.
Connection URL	Base URL to connect to the Log Analytics workspace in the following format: <pre>https://api.loganalytics.io/v1/workspaces/<workspace_id></pre> Where <workspace_id> is the ID of the Log Analytics workspace as described in step 5.a.ii.

Field	Description
	Note: This field is automatically set to the URL to connect to the Log Analytics workspace. Replace the <workspace_id> variable in the auto-generated URL with the workspace ID of your Log Analytics workspace.
Credential Information	
OAuth Client ID	Application (client) ID of your Azure client application as described in step 4.a.
OAuth Client Secret	Client Secret of your Azure client application as described in step 4.a.
OAuth Token URL	Token URL of your Azure client application. Based on the region of your Azure client application, enter the token URL in one of the following formats: - Global https://login.microsoftonline.com/<tenantid>/oauth2/v2.0/token - US Government https://login.microsoftonline.us/<tenantid>/oauth2/v2.0/token

Field	Description
	ntid>/oauth2/v2.0/token - China https://login.partner.microsoftonline.cn/<tenantid>/oauth2/v2.0/token - Germany https://login.microsoftonline.de/<tenantid>/oauth2/v2.0/token Where <tenantid> is the tenant ID of your Azure client application as described in step 4.a.

- e. Select **Configure and Get OAuth Token**.
- f. When the OAuth token flow is successfully completed, return to the Create connection for the software import task page using the back button for your browser.
- g. Set the Create or Edit connection task to complete by selecting **Mark as Complete**.
- c. Test the Log Analytics API connection to import software data from the Azure client application.
 - a. For the Test Connection task, select **Configure**.
 - b. Select the data source associated with the newly created connection in the **Name** column of the Data Sources list.
 - c. Select the **Test Load 20 Records** related link.
 - d. When the state changes to **Complete**, return to the setup by selecting **Back to Guided Setup** in the Help panel.

- e. Set the Test Connection task to complete by selecting **Mark as Complete**.
- d. Review the scheduled data imports configuration.
 - a. For the Set up scheduled import jobs task, select **Configure**.
 - b. Select the SG-Azure Software scheduled data import.
 - c. On the Scheduled Data Import form, verify the field values for the scheduled job and select the **Active** check box.

For more information, see [Schedule a data import](#).

- d. Select **Update**.
- e. Set the Set up scheduled import jobs task to complete by selecting **Mark as Complete** in the Help panel.
- 6. (Optional) Enable extended discovery using the run command.

The run command is used to run a set of commands on the VMs to fetch server information such as CPU count, cores, model, and manufacturer. For information about extended discovery using the run command, see the [Extended discovery with Run Command on Service Graph connector for Microsoft Azure \[KB2210811\]](#) article in the Now Support Knowledge Base.

- a. Download the scripts that are required to enable extended discovery.
 - a. For the Download the scripts task, select **Configure**.
 - b. After the scripts are downloaded, select **Mark as Complete** to set the Download the scripts task to complete.
- b. Configure an Azure storage account for extended discovery.
 - a. For the Configure Azure Storage account for Run Command task, select **Configure**.

For information about configuring an Azure storage account for extended discovery, see the [Instructions for setting up Run Command in Service Graph Connector for Azure \[KB2001668\]](#) article in the Now Support Knowledge Base.

- b. Set up a new Azure storage account.
 - c. Set up a container to upload the script files.
 - d. Set up another container to store the results.
 - e. Provide run command access to the service principal.
 - f. Select **Mark as Complete** to set the Configure Azure Storage account for Run Command task to complete.
- c. Configure the connection properties for the new instance that you created for Service Graph Connector for Microsoft Azure.
- a. For the Configure connection properties task, select **Configure**.
 - b. In the **Connection*** field of the SG-Azure Configuration Properties section, select the new instance that you created.

The fields in the Properties section are populated with the details of the instance.

- c. In the Properties section, provide the following details for the instance that you created:
- Name of the storage account where the containers are created
 - Subscription ID where the storage account is created
 - Resource group of the storage account
 - Name of the container where the sh and ps1 files are uploaded in Azure
 - Name of the container where the results of the commands are to be stored
 - URI of the sh file that has to be run on Linux machines
 - URI of the ps1 file that has to be run on Windows machines
- d. Select **Save**.

- e. Select **Mark as Complete** to set the Configure connection properties task to complete.

Result

The data from Azure is pulled into your CMDB.

CMDB classes targeted in Service Graph Connector for Microsoft Azure

When you complete setting up the connection, you can configure the integration to periodically pull data from Microsoft Azure. The data is saved in tables that extend from the Configuration item [cmdb_ci] table.

Application [cmdb_ci_appl]

The following attributes in the Application [cmdb_ci_appl] table are populated by collected data:

Attribute label	Attribute name
Class	sys_class_name
Version	version
TCP port(s)	tcp_port
Running Process	running_process
Running process key parameters	running_process_key_parameters
Name	name
Running process command	running_process_command

Relationship created for Application

Parent class	Relationship type	Child class
Application [cmdb_ci_appl]	Runs on::Runs	Computer [cmdb_ci_computer]

Availability Zone [cmdb_ci_availability_zone]

The following attributes in the Availability Zone [cmdb_ci_availability_zone] table are populated by collected data.

Attribute label	Attribute name
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationship created for Availability Zone

Parent class	Relationship type	Child class
Availability Zone [cmdb_ci_availability_zone]	Reference	Key Value [cmdb_key_value]

Azure Datacenter [cmdb_ci_azure_datacenter]

The following attributes in the Azure Datacenter [cmdb_ci_azure_datacenter] table are populated by collected data.

Attribute label	Attribute name
Comments	comments
Name	name
Object ID	object_id
Region	region
Status	install_status

Relationships created for Azure Datacenter

Parent class	Relationship type	Child class
Azure Datacenter [cmdb_ci_azure_datacenter]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Azure Datacenter [cmdb_ci_azure_datacenter]	Contains::Contained by	Resource Group [cmdb_ci_resource_group]
Azure Datacenter [cmdb_ci_azure_datacenter]	Contains::Contained by	Availability Zone [cmdb_ci_availability_zone]

Cloud DataBase [cmdb_ci_cloud_database]

The following attributes in the Cloud DataBase [cmdb_ci_cloud_database] table are populated by collected data.

Attribute label	Attribute name
Fully qualified domain name	fqdn
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state
Type	type
Version	version

Relationships created for Cloud DataBase

Parent class	Relationship type	Child class
Cloud DataBase [cmdb_ci_cloud_data_base]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_data_center]
Cloud DataBase [cmdb_ci_cloud_data_base]	Reference	Key Value [cmdb_key_value]

Cloud Function [cmdb_ci_cloud_function]

The following attributes in the Cloud Function [cmdb_ci_cloud_function] table are populated by collected data.

Attribute label	Attribute name
App Function state	app_function_state
Code Location URL	code_location_url
Fully qualified domain name	fqdn
Function Last Modified	function_last_modified
Install Status	install_status
IP Address	ip_address
Name	name
Object ID	object_id
Operational status	operational_status

Relationship created for Cloud Function

Parent class	Relationship type	Child class
Cloud Function [cmdb_ci_cloud_function]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]

Cloud LB IPAddress [cmdb_ci_cloud_lb_ipaddress]

The following attributes in the Cloud LB IPAddress [cmdb_ci_cloud_lb_ipaddress] table are populated by collected data.

Attribute label	Attribute name
Fully qualified domain name	fqdn
Install Status	install_status
IP Address	ip_address
IPAddress Type	ipaddress_type
Name	name
Object ID	object_id
Operational status	operational_status

Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

The following attributes in the Cloud Load Balancer [cmdb_ci_cloud_load_balancer] table are populated by collected data.

Attribute label	Attribute name
Canonical Hosted Zone Name	canonical_hosted_zone_name
DNS Name	dns_name
Fully qualified domain name	fqdn

Attribute label	Attribute name
Install Status	install_status
IP Address	ip_address
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationships created for Cloud Load Balancer

Parent class	Relationship type	Child class
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_data_center]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Owns::Owned by	Cloud LB IPAddress [cmdb_ci_cloud_lb_ip_address]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Reference	Key Value [cmdb_key_value]

Cloud Mgmt Network Interface [cmdb_ci_nic]

The following attributes in the Cloud Mgmt Network Interface [cmdb_ci_nic] table are populated by collected data.

Attribute label	Attribute name
Configuration Item	cmdb_ci
Install Status	install_status
IP Address	ip_address

Attribute label	Attribute name
MAC Address	mac_address
Name	name
Object ID	object_id
Operational status	operational_status
Primary	primary
Private IP	private_ip
Public DNS	public_dns
Public IP	public_ip
State	state
Static	is_static

Relationships created for Cloud Mgmt Network Interface

Parent class	Relationship type	Child class
Cloud Mgmt Network Interface [cmdb_ci_nic]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Contains::Contained by	Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Reference	Key Value [cmdb_key_value]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Reference	Virtual Machine Instance

Parent class	Relationship type	Child class
		[cmdb_ci_vm_instance]

Cloud Network [cmdb_ci_network]

The following attributes in the Cloud Network [cmdb_ci_network] table are populated by collected data.

Attribute label	Attribute name
Cidr	cidr
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationships created for Cloud Network

Parent class	Relationship type	Child class
Cloud Network [cmdb_ci_network]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Compute Security Group [cmdb_ci_compute_security_group]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]

Parent class	Relationship type	Child class
Cloud Network [cmdb_ci_network]	Reference	Key Value [cmdb_key_value]

Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]

The following attributes in the Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress] table are populated by collected data.

Attribute label	Attribute name
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
Public DNS	public_dns
Public IP Address	public_ip_address
State	state

Relationships created for Cloud Public IP Address

Parent class	Relationship type	Child class
Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]	Reference	Key Value [cmdb_key_value]

Cloud Resource [cmdb_ci_cmp_resource]

The following attributes in the Cloud Resource [cmdb_ci_cmp_resource] table are populated by collected data.

Attribute label	Attribute name
Description	short_description
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
Resource type	resource_type
State	state

Relationships created for Cloud Resource

Parent class	Relationship type	Child class
Cloud Resource [cmdb_ci_cmp_resource]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_data_center]
Cloud Resource [cmdb_ci_cmp_resource]	Reference	Key Value [cmdb_key_value]

Cloud Service Account [cmdb_ci_cloud_service_account]

The following attributes in the Cloud Service Account [cmdb_ci_cloud_service_account] table are populated by collected data.

Attribute label	Attribute name
Account Id	account_id

Attribute label	Attribute name
Name	name
Object ID	object_id
Datacenter Type	datacenter_type

Relationship created for Cloud Service Account

Parent class	Relationship type	Child class
Cloud Service Account [cmdb_ci_cloud_service_account]	Reference	Key Value [cmdb_key_value]

Cloud Storage Account [cmdb_ci_cloud_storage_account]

The following attributes in the Cloud Storage Account [cmdb_ci_cloud_storage_account] table are populated by collected data.

Attribute label	Attribute name
Blob Service	blob_service
File Service	file_service
Install Status	install_status
Name	name
Object ID	object_id
Queue Service	queue_service
Sku Name	sku_name
State	state
Table Service	table_service

Relationships created for Cloud Storage Account

Parent class	Relationship type	Child class
Cloud Storage Account [cmdb_ci_cloud_storage_account]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Cloud Storage Account [cmdb_ci_cloud_storage_account]	Reference	Key Value [cmdb_key_value]

Cloud Subnet [cmdb_ci_cloud_subnet]

The following attributes in the Cloud Subnet [cmdb_ci_cloud_subnet] table are populated by collected data.

Attribute label	Attribute name
CIDR	cidr
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationship created for Cloud Subnet

Parent class	Relationship type	Child class
Cloud Subnet [cmdb_ci_cloud_subnet]	Reference	Key Value [cmdb_key_value]

Compute Security Group [cmdb_ci_compute_security_group]

The following attributes in the Compute Security Group [cmdb_ci_compute_security_group] table are populated by collected data.

Attribute label	Attribute name
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationships created for Compute Security Group

Parent class	Relationship type	Child class
Compute Security Group [cmdb_ci_compute_security_group]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Compute Security Group [cmdb_ci_compute_security_group]	Reference	Key Value [cmdb_key_value]

Computer [cmdb_ci_computer]

The following attributes in the Computer [cmdb_ci_computer] table are populated by collected data.

Attribute label	Attribute name
Name	name
CPU core count	cpu_core_count

Attribute label	Attribute name
CPU count	cpu_count
Disk space (GB)	disk_space
IP Address	ip_address
Is Virtual	virtual
Object ID	object_id
Operating System	os
Operational status	operational_status
OS Version	os_version
RAM (MB)	ram
Serial number	serial_number

Relationships created for Computer

Parent class	Relationship type	Child class
Computer [cmdb_ci_computer]	Owns::Owned by	IP Address [cmdb_ci_ip_address]
Computer [cmdb_ci_computer]	Virtualized by::Virtualizes	Virtual Machine Instance [cmdb_ci_vm_instance]

For information about the classification of VDIs into the Computer [cmdb_ci_computer] CI class, see the [Classification of VDIs into computer class using Service graph connector for Azure \[KB2184443\]](#) article in the Now Support Knowledge Base.

Hardware Type [cmdb_ci_compute_template]

The following attributes in the Hardware Type [cmdb_ci_compute_template] table are populated by collected data.

Attribute label	Attribute name
Name	name
Object ID	object_id
Class	sys_class_name
Cores	cores
Logical Storage GB	local_storage_gb
Memory MB	memory_mb
vCPUs	vcpus

Relationship created for Hardware Type

Parent class	Relationship type	Child class
Hardware Type [cmdb_ci_compute_t emplate]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_data center]

Note: When the Cloud Hardware Type class extension is enabled, the **Class** attribute is set to **Cloud Hardware Type**. Else, the attribute is set to **Hardware Type**.

As a user with the admin role, you can enable the Cloud Hardware Type class extension by setting the **use a single hardware type for cloud data centers** property (sn_itom_pattern.use a single hardware type for cloud data centers) to **true**. For more information, see the [Service Graph Connector For Microsoft Azure - Migrating to a new hardware type model \[KB1288455\]](#) article in the Now Support Knowledge Base.

Relationship created for Hardware Type

Parent class	Relationship type	Child class
Hardware Type [cmdb_ci_compute_template]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]

Image [cmdb_ci_os_template]

The following attributes in the Image [cmdb_ci_os_template] table are populated by collected data.

Attribute label	Attribute name
Guest OS	guest_os
Image Source	image_source
Install Status	install_status
Name	name
Object ID	object_id
Offer	offer
Operational status	operational_status
Version	version
Vendor	vendor

Relationship created for Image

Parent class	Relationship type	Child class
Image [cmdb_ci_os_template]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]

Instance Scale Set [cmdb_ci_instance_scale_set]

The following attributes in the Instance Scale Set [cmdb_ci_instance_scale_set] table are populated by collected data.

Attribute label	Attribute name
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Relationship created for Instance Scale Set

Parent class	Relationship type	Child class
Instance Scale Set [cmdb_ci_instance_scale_set]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]

IP Address [cmdb_ci_ip_address]

The following attributes in the IP Address [cmdb_ci_ip_address] table are populated by collected data.

Attribute label	Attribute name
Owned By Configuration Item	owned_by_cmdb_ci
IP Address	ip_address
Name	name
Nic	nic

Relationships created for IP Address

Parent class	Relationship type	Child class
IP Address [cmdb_ci_ip_address]	Reference	Cloud Mgmt Network Interface [cmdb_ci_nic]
IP Address [cmdb_ci_ip_address]	Reference	Computer [cmdb_ci_computer]

Key Value [cmdb_key_value]

The following attributes in the Key Value [cmdb_key_value] table are populated by collected data.

Attribute label	Attribute name
Key	key
Value	value

Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

The following attributes in the Kubernetes Cluster [cmdb_ci_kubernetes_cluster] table are populated by collected data.

Attribute label	Attribute name
Cluster Resource ID	cluster_resource_id
Cluster Version	cluster_version
Fully qualified domain name	fqdn
Install Status	install_status
Kubernetes UID	k8s_uid
Name	name

Relationship created for Kubernetes Cluster

Parent class	Relationship type	Child class
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]

Linux Server [cmdb_ci_linux_server]

The following attributes in the Linux Server [cmdb_ci_linux_server] table are populated by collected data.

Attribute label	Attribute name
CPU core count	cpu_core_count
Disk space (GB)	disk_space
Install Status	install_status
Is Virtual	virtual
Name	name
Object ID	object_id
Operating System	os
Operational status	operational_status
OS Version	os_version
RAM (MB)	ram
Serial number	serial_number

Relationship created for Linux Server

Parent class	Relationship type	Child class
Linux Server [cmdb_ci_linux_server]	Virtualized by::Virtualizes	Virtual Machine Instance [cmdb_ci_vm_instance]

Resource Group [cmdb_ci_resource_group]

The following attributes in the Resource Group [cmdb_ci_resource_group] table are populated by collected data.

Attribute label	Attribute name
Name	name
Object ID	object_id
Operational status	operational_status
Install Status	install_status
State	state

Relationships created for Resource Group

Parent class	Relationship type	Child class
Resource Group [cmdb_ci_resource_group]	Contains::Contained by	Virtual Machine Instance [cmdb_ci_vm_instance]
Resource Group [cmdb_ci_resource_group]	Contains::Contained by	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]
Resource Group [cmdb_ci_resource_group]	Contains::Contained by	Instance Scale Set [cmdb_ci_instance_scale_set]

Parent class	Relationship type	Child class
Resource Group [cmdb_ci_resource_group]	Contains::Contained by	Image [cmdb_ci_os_template]
Resource Group [cmdb_ci_resource_group]	Contains::Contained by	Storage Volume [cmdb_ci_storage_volume]
Resource Group [cmdb_ci_resource_group]	Reference	Key Value [cmdb_key_value]

Serial Number [cmdb_serial_number]

The following attributes in the Serial Number [cmdb_serial_number] table are populated by collected data.

Attribute label	Attribute name
Serial Number	serial_number
Serial Number Type	serial_number_type
Valid	valid

Relationships created for Serial Number

Parent class	Relationship type	Child class
Serial Number [cmdb_serial_number]	Reference	Server [cmdb_ci_server]
Serial Number [cmdb_serial_number]	Reference	Windows Server [cmdb_ci_win_server]
Serial Number [cmdb_serial_number]	Reference	Linux Server [cmdb_ci_linux_server]

Server [cmdb_ci_server]

The following attributes in the Server [cmdb_ci_server] table are populated by collected data.

Attribute label	Attribute name
CPU core count	cpu_core_count
CPU count	cpu_count
Disk space (GB)	disk_space
Install Status	install_status
Is Virtual	virtual
Name	name
Object ID	object_id
Operating System	os
Operational status	operational_status
OS Version	os_version
RAM (MB)	ram
Serial number	serial_number
Comments	comments

Relationships created for Server

Parent class	Relationship type	Child class
Server [cmdb_ci_server]	Owns::Owned by	Cloud Mgmt Network Interface [cmdb_ci_nic]

Parent class	Relationship type	Child class
Server [cmdb_ci_server]	Virtualized by::Virtualizes	Virtual Machine Instance [cmdb_ci_vm_instanc e]

Software [cmdb_ci_spkg]

The following attributes in the Software [cmdb_ci_spkg] table are populated by collected data.

Attribute label	Attribute name
Key	key
Name	name
Version	version

Relationship created for Software

Parent class	Relationship type	Child class
Software [cmdb_ci_spkg]	Reference	Software Instance [cmdb_software_insta nce]

Software Installation [cmdb_sam_sw_install]

The following attributes in the Software Installation [cmdb_sam_sw_install] table are populated by collected data.

Attribute label	Attribute name
Display name	display_name
Publisher	publisher
Version	version

Attribute label	Attribute name
Discovery source	discovery_source

Software Instance [cmdb_software_instance]

The following attributes in the Software Instance [cmdb_software_instance] table are populated by collected data.

Attribute label	Attribute name
Installed on	installed_on
Name	name

Relationships created for Software Instance

Parent class	Relationship type	Child class
Software Instance [cmdb_software_instance]	Reference	Computer [cmdb_ci_computer]

Storage Volume [cmdb_ci_storage_volume]

The following attributes in the Storage Volume [cmdb_ci_storage_volume] table are populated by collected data.

Attribute label	Attribute name
Install Status	install_status
Name	name
Object ID	object_id
Operational status	operational_status
Size	size
Size bytes	size_bytes

Attribute label	Attribute name
State	state
Volume ID	volume_id

Relationships created for Storage Volume

Parent class	Relationship type	Child class
Storage Volume [cmdb_ci_storage_volume]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Storage Volume [cmdb_ci_storage_volume]	Reference	Key Value [cmdb_key_value]

Virtual Machine Instance [cmdb_ci_vm_instance]

The following attributes in the Virtual Machine Instance [cmdb_ci_vm_instance] table are populated by collected data.

Attribute label	Attribute name
CPUs	cpus
Disks size (GB)	disks_size
Install Status	install_status
IP Address	ip_address
Memory (MB)	memory
Name	name
Object ID	object_id
Operational status	operational_status
State	state

Attribute label	Attribute name
VM Instance ID	vm_inst_id

Relationships created for Virtual Machine Instance

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Managed by::Manages	Instance Scale Set [cmdb_ci_instance_scale_set]
Virtual Machine Instance [cmdb_ci_vm_instance]	Hosted on::Hosts	Azure Datacenter [cmdb_ci_azure_datacenter]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Image [cmdb_ci_os_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Hardware Type [cmdb_ci_compute_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Contains::Contained by	Cloud Mgmt Network Interface [cmdb_ci_nic]
Virtual Machine Instance [cmdb_ci_vm_instance]	Uses::Used by	Storage Volume [cmdb_ci_storage_volume]
Virtual Machine Instance [cmdb_ci_vm_instance]	Reference	Key Value [cmdb_key_value]

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Reference	Computer [cmdb_ci_computer]

Windows Server [cmdb_ci_win_server]

The following attributes in the Windows Server [cmdb_ci_win_server] table are populated by collected data.

Attribute label	Attribute name
CPU core count	cpu_core_count
Disk space (GB)	disk_space
Install Status	install_status
Is Virtual	virtual
Name	name
Object ID	object_id
Operating System	os
Operational status	operational_status
OS Version	os_version
RAM (MB)	ram
Serial number	serial_number

Relationship created for Windows Server

Parent class	Relationship type	Child class
Windows Server [cmdb_ci_win_server]	Virtualized by::Virtualizes	Virtual Machine Instance

Parent class	Relationship type	Child class
		[cmdb_ci_vm_instance]

Quick start tests for Service Graph Connector for Microsoft Azure

Validate that Service Graph Connector for Microsoft Azure still works after you make any configuration change such as applying an upgrade or developing an application. Copy and customize these quick start tests to pass when using your instance-specific data.

Danger: By default, the system property that is used to run automated tests is disabled to prevent you from accidentally running these tests on a production system. To avoid data corruption or an outage, run tests only on development, test, and other non-production instances. See [Enable or disable executing Automated Test Framework tests](#).

Service Graph Connector for Microsoft Azure

SG-Azure: Service Graph Connector for Microsoft Azure Validation test suite

Test suite to verify the integrity of an integration using multiple tests.

Test	Description	Release version
SG-Azure Availability Zone	Validate the successful ingestion of the record in the Availability Zone table with the prerequisite parent ingestion.	Utah
SG-Azure Data Center	Validate the successful ingestion of multiple records in the Azure Data	Utah

Test	Description	Release version
	Center table with the prerequisite parent ingestion.	
SG-Azure Functions	Validate the successful ingestion of multiple records in the Cloud Service Account, Data Center, Cloud Database, and Cloud Function.	Utah
SG-Azure Generic Resources	Validate the successful ingestion of multiple records in the Cloud Resource and Relations table.	Utah
SG-Azure Hardware Template	Validate the successful ingestion of multiple records in the Hardware Template and Compute Template tables.	Utah
SG-Azure Load Balancers	Validate the successful ingestion of multiple records in the Load Balancer, Computer Security Group and Relations table with the prerequisite parent ingestion.	Utah
SG-Azure Network	Validate the successful ingestion of multiple records in the Network table with	Utah

Test	Description	Release version
	the prerequisite parent ingestion.	
SG-Azure Network Interface	Validate the successful ingestion of multiple records in the Networks, Public IP Address and Relations table.	Utah
SG-Azure Public IP Address	Validate the successful ingestion of multiple records in the Public IP Address, Networks, and Load Balancer tables.	Utah
SG-Azure Resource Group	Validate the successful ingestion of records in the Resource Group table.	Utah
SG-Azure Scale Sets	Validate the successful ingestion of records in the Scale Sets and Relations tables.	Utah
SG-Azure Scale Sets VMs	Validate the successful ingestion of multiple records in the Servers, Virtual Machine Instance, Scale Sets, and Relations tables with the prerequisite parent ingestion.	Utah
SG-Azure Security Groups	Validate the successful ingestion of multiple records	Utah

Test	Description	Release version
	in the Cloud Service Account, Computer Security Group, and Relations tables with the prerequisite parent ingestion.	
SG-Azure Server Config Data	Validate the successful ingestion of multiple records in the Virtual Machine Instance, Linux Server, and Relations tables with the prerequisite parent ingestion.	Utah
SG-Azure Software	Validate the successful ingestion of multiple records in the Virtual Machine Instance, Software Records, and Relations tables with the prerequisite parent ingestion.	Utah
SG-Azure SQL	Validate the successful ingestion of multiple records in the Data Center, Cloud Database, and Relations tables with the prerequisite parent ingestion.	Utah
SG-Azure Storage Account	Validate the successful ingestion of multiple records in the Storage Account, Computer Security Group and Relations	Utah

Test	Description	Release version
	tables with the prerequisite parent ingestion.	
SG-Azure Storage Volume	Validate the successful ingestion of multiple records in the Storage Volume, Computer Security Group and Relations tables with the prerequisite parent ingestion.	Utah
SG-Azure Subscription	Validate the successful ingestion of records in the Cloud Service Account table.	Utah
SG-Azure VMs	Validate the successful ingestion of multiple records in the Virtual Machines, Virtual Machine Instance and Relations tables with prerequisite parent ingestion.	Utah

To learn more about Service Graph Connector for Microsoft Azure, see [Service Graph Connector for Microsoft Azure](#).

Related topics

- [Quick start tests](#)

Accessing the connection details of Service Graph Connector for Microsoft Azure

You can access the connection details of the Service Graph Connector for Microsoft Azure in a single view using the common connection framework (CCF) included within the Integration Commons for CMDb (sn_cmdb_int_util) store app.

With the CCF, you can access all the connections used by the Service Graph Connector for Microsoft Azure. The connection details include the connection alias, connection properties, data sources, and scheduled data imports associated with a connection. You can also test the connection. For more information, see [Accessing the connection details of Service Graph Connectors](#).

Access the details of an Azure connection

Access the details of an Azure connection configured for the Service Graph Connector for Microsoft Azure.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > Azure > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. On the Service Graph Connections page, view the connection details such as the connection name, alias, and data sources.
4. Select a related list to view further details of the connection.
5. (Optional) Select the **Test Connection** related link to test the connection.

Additional features within the Service Graph Connector for Microsoft Azure

Use the additional features available within the Service Graph Connector for Microsoft Azure to maximize its use for importing data.

The following features are available within the Service Graph Connector for Microsoft Azure:

Configure multiple Azure connections

Follow the guided setup to establish a hardware or software connection, or configure a workspace when adding a software connection.

Set partition size for parallel loading-enabled Azure accounts

Optimize the performance of data sources by enabling parallel data loading.

Disable delta synchronization

Import full data from your Azure instance when needed.

Enabling VM insights for collecting Azure data

Enable the VM insights feature for all the VMs in your Log Analytics workspace to populate the VMConnection and VMProcess tables in Azure.

Setting up multiple connections in Service Graph Connector for Microsoft Azure

The Service Graph Connector for Microsoft Azure supports multiple Azure connections.

Important: Add multiple connections by selecting **Add connection** on the Connections page of the Workflow Studio in the guided setup. For more information, see [Configure Service Graph Connector for Microsoft Azure](#).

Setting up multiple connections

You can set up a hardware or software connection by using the guided setup. The Service Graph Connector for Microsoft Azure supports only one-to-one mapping between hardware and software connections. Therefore, when configuring multiple software connections, you must also configure an equivalent number of hardware connections. You can differentiate multiple connections using their respective connection aliases.

For example, if you add 10 software connections for 10 different workspaces, you must also add 10 hardware connections for each corresponding software connection.

Setting up multiple workspaces

You can set up a workspace when adding a software connection using the guided setup. To set up multiple workspaces, you must configure that many different software connections in the Service Graph Connector for Microsoft Azure as there are workspaces and an equivalent number of hardware connections.

For example, if you have 10 workspaces, you must add 10 software connections for each workspace and 10 hardware connections for each corresponding software connection.

Set partition size for parallel loading-enabled Azure accounts

Optimize the performance of data sources by setting the partition size in the `azure_full_pull_partition_size` system property.

Before you begin

Set the application scope to Service Graph Connector for Microsoft Azure using the application picker. For more information, see [Application picker](#).

Role required: admin

About this task

The parallel loading feature is available for the following data sources:

- Starting with the Service Graph Connector for Microsoft Azure 1.13.0 version:
 - SG-Azure Functions
 - SG-Azure Generic Resources
 - SG-Azure Get Run Command
 - SG-Azure Hardware Template Updation
 - SG-Azure HW Consolidation
 - SG-Azure Kubernetes Cluster
 - SG-Azure Load Balancers
 - SG-Azure Network Interface
 - SG-Azure Public IP Address
 - SG-Azure Run Command
 - SG-Azure Scale Sets VMs
- Starting with the Service Graph Connector for Microsoft Azure 1.11.0 version:
 - SG-Azure Network
 - SG-Azure Virtual Machines
 - SG-Azure Storage Volume

For a full data pull, the `azure_full_pull_partition_size` property determines the number of Azure subscriptions that can be executed as parallel jobs to retrieve configuration item (CI) data. Partition size determines how many Azure subscriptions a single parallel-thread evaluates for data retrieval. If you set the `azure_full_pull_partition_size` property to `100`, the partition size is 100 subscriptions.

Note: If the `azure_full_pull_partition_size` property isn't configured, the partition size computation feature of the Integration Commons for CMDb application automatically determines the partition size for parallel loading at runtime. See [Partition size computation for parallel loading in Integration Commons for CMDb](#).

Procedure

1. Navigate to **All > Service Graph Connectors > Azure > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. From the **Property** column of the Service Graph Connection Properties related list, select **azure_full_pull_partition_size**.
The `azure_full_pull_partition_size` system property determines the number of subscriptions that each partition processes for a full pull. For more information, see [Service Graph Connector for Microsoft Azure properties](#).
4. In the **Value** field, enter a numeric value greater than 0 for the partition size.
5. Select **Update**.

Note: After you enable parallel data loading for a data source, the **Test Load 20 Records** and **Load All Records** options don't work for executing the data source. You must select **Execute Now** from **Scheduled Data Import** to execute the data source for which parallel loading is enabled.

Configuring data synchronization in Service Graph Connector for Microsoft Azure

You can configure the system property for synchronizing only delta data from an Azure instance.

The connector provides the ability to import only delta data from an Azure instance. With delta synchronization enabled, the importing process of Azure data is quicker because the connector imports the changes in Azure data only.

For data sources discovered by a software connection, the **Last Run Datetime** field displays the timestamp of the most recent delta pull run. The **Last Run Datetime** field is updated only when the last delta pull run was a success. An empty **Last Run Datetime** field indicates no previous delta pull run for the data source, but either a full pull run or no run at all. The delta pull period is calculated by subtracting the last run date and time from the current date and time; that is, Delta period = Current date timestamp - Last run date timestamp.

For data sources discovered by a hardware connection, the **Last Success Import Time** field displays the timestamp of the most recent delta pull run. The **Last Success Import Time** field is updated only when the last delta pull run was a success. An empty **Last Success Import Time** field indicates no previous delta pull run for the data source, but either a full pull run or no run at all. The delta pull period is calculated by subtracting the last successful import date and time from the current date and time; that is, Delta period = Current date timestamp - Last success import date timestamp.

The application includes the `sn_sg_azure_integ.is_delta_sync_enabled` system property to enable delta synchronization. By default, the property is set to `true`. To import full data from your Azure instance, disable the `sn_sg_azure_integ.is_delta_sync_enabled` property. See [Disable delta synchronization](#).

For a delta pull, you can also configure the grace time, page size, preceding time period, and API call duration using system properties. For more information, see [Service Graph Connector for Microsoft Azure properties](#).

Disable delta synchronization

Disable delta synchronization to import all Microsoft Azure data.

Before you begin

Set the application scope to Service Graph Connector for Microsoft Azure using the application picker. For more information, see [Application picker](#).

Role required: admin

Procedure

1. Enter `sys_properties.list` in the navigation filter, and then open the `sn_sg_azure_integ.is_delta_sync_enabled` property.
2. In the **Value** field, enter `false`.
3. Select **Update**.

Enabling VM insights for collecting Azure data

You must enable the VM insights feature in the Azure portal for collecting TCP and processes data using the Service Graph Connector for Microsoft Azure.

Requirements for enabling the VM insights feature

- Ensure that you've created the Log Analytics workspace to push the logs from virtual machines (VMs) via insights in Azure Monitor. See [Insights in Azure Monitor](#) on the Microsoft Azure documentation site.
- Note down the subscription name associated with your Log Analytics workspace.

Collecting the TCP and processes data

The SG-Azure TCP and SG-Azure VM Config Data data sources in the Service Graph Connector for Microsoft Azure import data from the VMConnection and VMProcess tables in Azure, respectively. To fetch data correctly from the VMConnection and VMProcess tables in Azure, you must enable the VM insights feature for all the VMs in your Log Analytics workspace. To learn how to populate these tables via insights, see [Enable VM insights in the Azure portal](#) on the Microsoft Azure documentation site.

There are several methods available for enabling VM insights. For more information, see [Enable VM insights overview](#) on the Microsoft Azure documentation site.

Note: In the data collection rule, ensure that you've selected the **Enable guest performance** and **Enable processes and dependencies (Maps)** check boxes on the rule form.

Service Graph Connector for Microsoft Azure reference

Reference topics provide additional information about Service Graph Connector for Microsoft Azure components.

- [Supported Microsoft Azure resource types](#)

Several Microsoft Azure resource types are imported as CMDB data.

- [Microsoft Azure discovery solutions comparison](#)

ITOM Visibility applications discover a variety of Microsoft Azure resources and populate the relevant configuration item (CI) classes in the Configuration Management Database (CMDB) with their attributes.

- [Life cycle management of records in Service Graph Connector for Microsoft Azure](#)

Life cycle management in the Service Graph Connector for Microsoft Azure monitors and updates the statuses of Azure resources throughout their entire life cycle, from creation to deletion.

- [Record removal process in Service Graph Connector for Microsoft Azure](#)

The Service Graph Connector for Microsoft Azure uses the Integration Commons record removal process for life cycle management during full data loads. For delta loads, life cycle management of records is based on updates from Microsoft Azure.

- [Service Graph Connector for Microsoft Azure properties](#)

Service Graph Connector for Microsoft Azure properties control the behavior of the connector.

Supported Microsoft Azure resource types

Several Microsoft Azure resource types are imported as CMDB data.

Microsoft Azure resource types refer to the names included in delta queries used for filtering data using the [Azure Resource Graph](#) API for a specific data source.

Note: Delta pulls include data from a specific time period that is defined by the condition within a delta query.

The following table lists the Service Graph Connector for Microsoft Azure data sources, Microsoft Azure resource types they import, and indicates whether they support both full pull and delta pull of data during import. ✓ indicates supported and ✗ indicates not supported.

Resource types supported in Service Graph Connector for Microsoft Azure

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
SG-Azure Subscriptions	Cloud Service Account [cmdb_ci_cloud_service_account]	Microsoft.Resources/subscriptions	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Datacenter Updation	Azure Datacenter [cmdb_ci_azure_datacenter]	None	✓	✓
SG-Azure Resource Group	Resource Group [cmdb_ci_resource_group]	Microsoft.Resources/subscriptions/resourcegroups	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Availability Zone	Availability Zone [cmdb_ci_availability_zone]	Microsoft.Compute/availabilitySets	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Storage Volume	Storage Volume [cmdb_ci_storage_volume]	Microsoft.Compute/disks	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Load Balancers	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Microsoft.Network/loadBalancers	✓	✓
	Cloud LB IPAddress [cmdb_ci_cloud_lb_ipaddress]	Microsoft.Network/loadBalancers	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Network	Cloud Network [cmdb_ci_network]	Microsoft.Network/virtualnetworks	✓	✓
	Cloud Service Account [cmdb_ci_cloud_service_account]	Microsoft.Network/virtualnetworks	✓	✓
	Cloud Subnet [cmdb_ci_cloud_subnet]	Microsoft.Network/virtualnetworks	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Network Interface	Cloud Mgmt Network Interface [cmdb_ci_nic]	Microsoft.Network/networkInterfaces	✓	✓
	IP Address [cmdb_ci_ip_address]	Microsoft.Network/networkInterfaces	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Hardware [cmdb_ci_hardware]	Microsoft.Network/networkInterfaces	✓	✓
	Virtual Machine Instance [cmdb_ci_vm_instance]	Microsoft.Network/networkInterfaces	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Public IP Address	Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]	Microsoft.Network/publicIPAddresses	✓	✓
	Cloud Mgmt Network Interface [cmdb_ci_nic]	Microsoft.Network/publicIPAddresses	✓	✓
	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Microsoft.Network/publicIPAddresses	✓	✓
	Cloud LB IPAddress [cmdb_ci_cloud_lb_ipaddress]	Microsoft.Network/publicIPAddresses	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Security Group	Compute Security Group [cmdb_ci_compute_security_group]	Microsoft.Network/networkSecurityGroups	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Storage Accounts	Cloud Storage Account [cmdb_ci_cloud_storage_account]	Microsoft.Storage/storageAccounts	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Virtual Machines	Virtual Machine Instance [cmdb_ci_vm_instance]	Microsoft.Compute/virtualMachines	✓	✓
	Image [cmdb_ci_os_template]	Microsoft.Compute/virtualMachines	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Linux Server [cmdb_ci_linux_server]	Microsoft.Compute/virtualMachines	✓	✓
	Server [cmdb_ci_server]	Microsoft.Compute/virtualMachines	✓	✓
	Windows Server [cmdb_ci_win_server]	Microsoft.Compute/virtualMachines	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Software	When the Software Asset Management (SAM) application isn't installed: Software [cmdb_ci_spkg] Software Installation [cmdb_sam_sw_install]	None	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	When the SAM application is installed: Software Instance [cmdb_software_instance]	None	✓	✓
SG-Azure SQL	Cloud DataBase [cmdb_ci_cloud_database]	Microsoft.Sql/servers Microsoft.DBforPostgreSQL/servers microsoft.dbformysql/flexible servers microsoft.dbforpostgresql/flexible servers	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure TCP	TCP Connection [cmdb_tcp]	None	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Running Process [cmdb_running_process]	None	✓	✓
SG-Azure Functions	Cloud Function [cmdb_ci_cloud_function]	Microsoft.Web/sites	✓	✓
SG-Azure Generic Resources	Cloud Resource [cmdb_ci_compute_resource]	Microsoft.ResourceGraph/resources - Generic	✓	✓
	Key Value [cmdb_key_value]	Tags	✓	✓
SG-Azure Scale Sets	Instance Scale Set [cmdb_ci_instance_scale_set]	Microsoft.Compute/virtualmachinescalesets	✓	✓
SG-Azure Scale Sets VMs Note: Imports uniform orchestration scale sets VMs.	Virtual Machine Instance [cmdb_ci_vm_instance]	None	✓	✗
	Storage Volume [cmdb_ci_storage_volume]	None	✓	✗

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	Image [cmdb_ci_os_template]	None	✓	×
	Cloud Mgmt Network Interface [cmdb_ci_nic]	None	✓	×
	Hardware Type [cmdb_ci_compute_template]	None	✓	×
	Azure Datacenter [cmdb_ci_azure_datacenter]	None	✓	×
	Linux Server [cmdb_ci_linux_server]	None	✓	×
	Server [cmdb_ci_server]	None	✓	×
	Windows Server [cmdb_ci_windows_server]	None	✓	×
	Key Value [cmdb_key_value]	Tags	✓	×

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
SG-Azure Software Remove	When the SAM application isn't installed: Software [cmdb_ci_spkg] Software Installation [cmdb_sam_sw_install]	None	✓	✓
	When the SAM application is installed: Software Instance [cmdb_software_instance]	None	✓	✓
SG-Azure Hardware Template Updation	Hardware Type [cmdb_ci_compute_template]	None	✓	✓
SG-Azure HW Consolidation	Virtual Machine Instance	None	✓	✓

Data source	CMDB CI classes	Microsoft Azure resource type	Full pull	Delta pull
	[cmdb_ci_vm_instance]			
	Computer [cmdb_ci_computer]	None	✓	✓
SG-Azure Run Command	None	None	✓	✗
	Application [cmdb_ci_app]	None	✓	✗
	TCP Connection [cmdb_tcp]	None	✓	✗
	Running Process [cmdb_running_process]	None	✓	✗
SG-Azure Kubernetes Cluster	SG-Azure Kubernetes Cluster [sn_sg_azure_integ_kubernetes_cluster]	microsoft.containerservice/managedclusters	✓	✓

Microsoft Azure discovery solutions comparison

ITOM Visibility applications discover a variety of Microsoft Azure resources and populate the relevant configuration item (CI) classes in the Configuration Management Database (CMDB) with their attributes.

ITOM Visibility solutions

ITOM Visibility can offer you comprehensive visibility into your organization's cloud infrastructure. Each application offers unique strengths and capabilities, and by comparing them you can gain holistic insights on their capabilities and benefits.

ITOM Visibility solutions for discovering Azure

ITOM Visibility solution	Overview	Configuration
Discovery and Service Mapping Patterns and Visibility Content	Discovery patterns used by ITOM Visibility	- You must update your store apps to the latest release version. - You must provide ITOM Visibility with access permissions. For more information, see Set up Azure service accounts
Service Graph Connector for Microsoft Azure	Service Graph Connector for Microsoft Azure	Configure Service Graph Connector for Microsoft Azure using the guided setup

Note:

The CI class, in most cases, is applicable to both solutions. If it's not applicable to one of the solutions, the solution is marked "N/A".

Discovery and Service Mapping Patterns vs Service Graph Connector for Microsoft Azure

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
microsoft.compute/availabilitysets	Yes	N/A	cmdb_ci_availability_set
microsoft.compute/availabilitysets	Yes	Yes	cmdb_ci_availability_zone
Microsoft.Management/managementGroups	Yes	No	cmdb_ci_azure_management_group
microsoft.KeyVault/vaults	Yes	No	cmdb_ci_certificate
- Microsoft.Sql/servers/databases - Microsoft.DBforMySQL/servers - Microsoft.DocumentDB/databaseAccounts - Microsoft.Cache/Redis	Yes	No	cmdb_ci_cloud_database

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
- Microsoft.Sql/servers - Microsoft.DBfor PostgreSQL/servers	Yes	Yes	cmdb_ci_cloud_database
microsoft.network/expressroutecircuits	Yes	No	cmdb_ci_cloud_direct_connect
Microsoft.Storage/storageAccounts/fileServices	Yes	No	cmdb_ci_cloud_file_service
Microsoft.Storage/storageAccounts/fileServices/shares	Yes	No	cmdb_ci_cloud_file_share
Microsoft.Web/sites/functions	Yes	Yes	cmdb_ci_cloud_function
VM Size	Yes	No	cmdb_ci_cloud_hardware_type
microsoft.compute/hostgroups/hosts	Yes	No	cmdb_ci_cloud_host
microsoft.network/	Yes	No	cmdb_ci_cloud_load_balancer

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
applicationgateways			
microsoft.networkk/loadbalancers	Yes	Yes	cmdb_ci_cloud_load_balancer
Microsoft.Storagee/storageAccounts/blobService	Yes	No	cmdb_ci_cloud_object_service
Tenant	Yes	No	cmdb_ci_cloud_org
microsoft.networkk/publicipaddresses	Yes	Yes	cmdb_ci_cloud_public_ipaddresses
Microsoft.Storagee/storageAccounts/queueServices	Yes	No	cmdb_ci_cloud_queue_service
Microsoft.Resources/subscriptions	Yes	Yes	cmdb_ci_cloud_service_account
Microsoft.Management/managementGroup	Yes	No	cmdb_ci_cloud_service_account
microsoft.storagee/storageaccounts	Yes	Yes	cmdb_ci_cloud_storage_account

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
Microsoft.Network/ virtualNetworks/ subnets	Yes	No	cmdb_ci_cloud_subnet
Microsoft.Storage/ storageAccounts/ tableServices	Yes	No	cmdb_ci_cloud_table_service
microsoft.web/ sites	Yes	No	cmdb_ci_cloud_webserver
microsoft.network/ networksecurity groups	Yes	Yes	cmdb_ci_compute_security_group
microsoft.network/ privatednszones	Yes	No	cmdb_ci_dns_zone
microsoft.compute/ virtualmachines calesets	Yes	Yes	cmdb_ci_instance_scale_set
microsoft.containerservice/ managedclusters	Yes	Yes	cmdb_ci_kubernetes_cluster
Microsoft.Network/ loadBalancers/ backendAddressPools	Yes	No	cmdb_ci_lb_pool

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
Microsoft.Network/loadBalancers/inboundNatRules	Yes	No	cmdb_ci_lb_service
microsoft.networkk/natgateways	Yes	No	cmdb_ci_nat_gateway
microsoft.networkk/virtualnetworks	Yes	Yes	cmdb_ci_network
microsoft.networkk/networkinterfaces	Yes	Yes	cmdb_ci_nic
microsoft.compute/images	Yes	No	cmdb_ci_os_template
Microsoft.Resources/resourceGroups	Yes	Yes	cmdb_ci_resource_group
Microsoft.Network/routeTables/routes	Yes	No	cmdb_ci_route_table
microsoft.networkk/routetables	Yes	No	cmdb_ci_route_table
microsoft.compute/disks	Yes	Yes	cmdb_ci_storage_volume
microsoft.networkk/	Yes	No	cmdb_ci_virtual_pvt_gateway

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
localnetworkgateways			
microsoft.network/virtualnetworkgateways	Yes	No	cmdb_ci_virtual_pvt_gateway
microsoft.compute/virtualmachines	Yes	Yes	cmdb_ci_vm_instance
microsoft.sqlvirtualmachine/sqlvirtualmachines	Yes	No	cmdb_ci_vm_instance
Microsoft.Network/virtualNetworks/virtualNetworkPeerings	Yes	No	cmdb_ci_vnet_peering
microsoft.network/connections	Yes	No	cmdb_ci_vpc_gateway_connection
Microsoft.Resources/tags	Yes	Yes	cmdb_key_value
Microsoft.Compute/virtualMachines	No	Yes	cmdb_ci_server
Microsoft.Compute/virtualMachines	No	Yes	cmdb_ci_win_server

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
Microsoft.Network/networkInterfaces	No	Yes	cmdb_ci_hardware
Microsoft.Compute/virtualMachines	No	Yes	cmdb_ci_linux_server
microsoft.containerservice/managedclusters	Yes	Yes	cmdb_ci_kubernetes_cluster
Microsoft.Compute/virtualmachinescalesets	No	Yes	cmdb_ci_vm_instance
Microsoft.ServiceBus/namespaces/queues	Yes	No	cmdb_azure_service_bus_queue
Microsoft.ServiceBus/namespaces/topics	Yes	No	cmdb_azure_service_bus_topic
microsoft.appconfiguration/configurationstores	Yes	No	cmdb_azure_app_configuration_configuration_store
microsoft.automation/automationaccounts	Yes	No	cmdb_azure_automation_automation_account

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
microsoft.compute/diskencryptionsets	Yes	No	cmdb_azure_disk_encryption_set_disk_encryption_set
microsoft.compute/galleries/images	Yes	No	cmdb_azure_compute_gallery_image_definition
microsoft.compute/snapshots	Yes	No	cmdb_azure_compute_snapshot
microsoft.containerregistry/registries	Yes	No	cmdb_azure_container_registry_registry
microsoft.databricks/workspaces	Yes	No	cmdb_azure_databricks_workspace
microsoft.datafactory/factories	Yes	No	cmdb_azure_data_factory_data_factory
microsoft.dataprotection/backupvaults	Yes	No	cmdb_azure_data_protection_backup_vault
microsoft.eventgrid/systemtopics	Yes	No	cmdb_azure_event_grid_system_topic
microsoft.eventhub/namespaces	Yes	No	cmdb_azure_event_hub_namespace

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
microsoft.insights/components	Yes	No	cmdb_azure_application_insight_component
microsoft.network/applicationgateway/webapplicationfirewall/policies	Yes	No	cmdb_azure_web_application_firewall_policy
microsoft.operationalinsights/workspaces	Yes	No	cmdb_azure_log_analytics_workspace
microsoft.network/azurefirewalls	Yes	No	cmdb_azure_network_security_azure_firewall
microsoft.devicecenter/devcenters	Yes	No	cmdb_azure_dev_centers_dev_center
microsoft.kusto/clusters	Yes	No	cmdb_azure_data_explorer_clusters_data_explorer_clusters
microsoft.recoveryservices/vaults	Yes	No	cmdb_azure_recovery_services_vault
microsoft.recoveryservices/vaults/backupfabrics/protectioncontainers	Yes	No	cmdb_azure_recovery_services_vault_backup_item

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
iners/ protecteditems			
microsoft.dbforpostgresql/ servergroupsv2	Yes	No	cmdb_azure_cosmos_db_for_postgresql_cluster_cluster
microsoft.insights/ datacollectionrules	Yes	No	cmdb_azure_application_insight_data_collection_rule
microsoft.network/ serviceendpointpolicies	Yes	No	cmdb_azure_service_endpoint_policy_service_endpoint_policy

Life cycle management of records in Service Graph Connector for Microsoft Azure

Life cycle management in the Service Graph Connector for Microsoft Azure monitors and updates the statuses of Azure resources throughout their entire life cycle, from creation to deletion.

The life cycle management process helps maintain the accuracy and integrity of data in the Configuration Management Database (CMDB).

Life cycle management for CIs in Service Graph Connector for Microsoft Azure

The following table lists the configuration items (CIs) in CMDB and other non-CMDB tables for which life cycle management is available in Service Graph Connector for Microsoft Azure.

Supported tables for life cycle management

Data source	CMDB CI classes	Life cycle management available
SG-Azure Availability Zone	Availability Zone [cmdb_ci_availability_zone]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Datacenter Updation	Azure Datacenter [cmdb_ci_azure_data_center]	Not applicable
SG-Azure Functions	Cloud Function [cmdb_ci_cloud_function]	Yes
SG-Azure Generic Resources	Cloud Resource [cmdb_ci_cmp_resource]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Get Run Command	Application [cmdb_ci_appl]	Not applicable
	TCP Connection [cmdb_tcp]	Not applicable
	Running Process [cmdb_running_process]	Not applicable
SG-Azure Hardware Template Updation	Hardware Type [cmdb_ci_compute_template]	Not applicable
SG-Azure HW Consolidation	Virtual Machine Instance	Not applicable

Data source	CMDB CI classes	Life cycle management available
	[cmdb_ci_vm_instance]	
	Computer [cmdb_ci_computer]	Not applicable
SG-Azure Kubernetes Cluster	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Yes
SG-Azure Load Balancers	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Yes
	Cloud LB IPAddress [cmdb_ci_cloud_lb_ip_address]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Network	Cloud Network [cmdb_ci_network]	Yes
	Cloud Subnet [cmdb_ci_cloud_subnet]	Yes
	Cloud Service Account [cmdb_ci_cloud_service_account]	No
	Key Value [cmdb_key_value]	No
SG-Azure Network Interface	Cloud Mgmt Network Interface [cmdb_ci_nic]	Yes

Data source	CMDB CI classes	Life cycle management available
	IP Address [cmdb_ci_ip_address]	No
	Hardware [cmdb_ci_hardware]	Not applicable
	Virtual Machine Instance [cmdb_ci_vm_instance]	Not applicable
	Key Value [cmdb_key_value]	No
SG-Azure Public IP Address	Cloud Public IP Address [cmdb_ci_cloud_public_ipaddress]	Yes
	Cloud Mgmt Network Interface [cmdb_ci_nic]	Yes
	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Yes
	Cloud LB IPAddress [cmdb_ci_cloud_lb_ipaddress]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Resource Group	Resource Group [cmdb_ci_resource_group]	Yes

Data source	CMDB CI classes	Life cycle management available
	Key Value [cmdb_key_value]	No
SG-Azure Scale Sets	Instance Scale Set [cmdb_ci_instance_scale_set]	Yes
SG-Azure Scale Sets VMs	Virtual Machine Instance [cmdb_ci_vm_instance]	Yes
	Storage Volume [cmdb_ci_storage_volume]	Yes
	Image [cmdb_ci_os_template]	Not applicable
	Cloud Mgmt Network Interface [cmdb_ci_nic]	Yes
	Hardware Type [cmdb_ci_compute_template]	Not applicable
	Linux Server [cmdb_ci_linux_server]	Yes
	Server [cmdb_ci_server]	Yes
	Windows Server [cmdb_ci_win_server]	Yes
	Key Value [cmdb_key_value]	No

Data source	CMDB CI classes	Life cycle management available
SG-Azure Security Group	Compute Security Group [cmdb_ci_compute_security_group]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Software	When the Software Asset Management (SAM) application isn't installed: Software [cmdb_ci_spkg] Software Instance [cmdb_software_instance] When the SAM application is installed: Software Installation [cmdb_sam_sw_install]	Yes
SG-Azure Software Remove	When the SAM application isn't installed: Software [cmdb_ci_spkg] Software Instance [cmdb_software_instance]	Yes

Data source	CMDB CI classes	Life cycle management available
	When the SAM application is installed: Software Installation [cmdb_sam_sw_install]	
SG-Azure SQL	Cloud DataBase [cmdb_ci_cloud_data_base]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Storage Accounts	Cloud Storage Account [cmdb_ci_cloud_storage_account]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Storage Volume	Storage Volume [cmdb_ci_storage_volume]	Yes
	Key Value [cmdb_key_value]	No
SG-Azure Subscriptions	Cloud Service Account [cmdb_ci_cloud_service_account]	No
	Key Value [cmdb_key_value]	No
SG-Azure TCP	TCP Connection [cmdb_tcp]	No

Data source	CMDB CI classes	Life cycle management available
	Running Process [cmdb_running_process]	No
SG-Azure Virtual Machines	Linux Server [cmdb_ci_linux_server]	Yes
	Server [cmdb_ci_server]	Yes
	Windows Server [cmdb_ci_win_server]	Yes
	Image [cmdb_ci_os_template]	Yes
	Key Value [cmdb_key_value]	No
	Virtual Machine Instance [cmdb_ci_vm_instance]	Yes

Record removal process in Service Graph Connector for Microsoft Azure

The Service Graph Connector for Microsoft Azure uses the Integration Commons record removal process for life cycle management during full data loads. For delta loads, life cycle management of records is based on updates from Microsoft Azure.

Record removal is the process of handling data that is no longer needed. For any discovered resources deleted later after pulling data, the Service Graph Connector for Microsoft Azure automatically updates the **Install Status** field of the associated CMDB CI classes to indicate any retired or deleted records.

The record removal process in the Service Graph Connector for Microsoft Azure involves marking the **Install Status** of a record as **retired** rather than permanently deleting it from the system. The record remains in the database, making it possible to reference or restore it later. Record removal for Service Graph Connector for Microsoft Azure relies on the Source [sys_object_source table].

Starting with the Service Graph Connector for Microsoft Azure 1.14.0 version, life cycle management is supported for the following data sources in full data load and delta load:

- SG-Azure Kubernetes Cluster
- SG-Azure Scale Sets
- SG-Azure Generic Resources
- SG-Azure Functions
- SG-Azure SQL
- SG-Azure Network Interface
- SG-Azure Virtual Machines
- SG-Azure Storage Accounts
- SG-Azure Security Group
- SG-Azure Public IP Address
- SG-Azure Network
- SG-Azure Load Balancers
- SG-Azure Storage Volume
- SG-Azure Availability Zone
- SG-Azure Resource Group
- SG-Azure Subscriptions

When you run a full data load for the first time after upgrading to version 1.14.0, the Install Status of all the records that aren't discovered is set to retired.

Service Graph Connector for Microsoft Azure properties

Service Graph Connector for Microsoft Azure properties control the behavior of the connector.

System properties

These system properties are available for Service Graph Connector for Microsoft Azure.

Note: To open the System Properties [sys_properties] table, enter `sys_properties.list` in the navigation filter.

System properties for Service Graph Connector for Microsoft Azure

Property	Description
api_retry_max	Enter the maximum retry limit for a Microsoft Azure API call made from the data sources within the connector. - Type: integer - Default value: 3 - Location: System Property [sys_properties] table
api_retry_max_without_backoff	Enter the maximum retry limit for a Microsoft Azure API call made from the data sources within the connector, without using exponential backoff to increase the waiting time between consecutive retries. Type: integer

Property	Description
	• Default value: 3 • Location: System Property [sys_properties] table
azure_full_pull_partition_size	Set the partition size for Azure full data pull. This property determines the number of subscriptions that each partition processes. The value must be set to a numeric value greater than 0. • Type: integer • Default value: null • Location: System Property [sys_properties] table Note: If no value is specified, the property value is determined based on the number of active threads in the instance. See Set partition size for parallel loading-enabled Azure accounts.
delta_grace_time	Enter the additional time period in hours added to the current delta pull period when fetching incremental changes. • Type: integer • Default value: 0

Property	Description
	Location: System Property [sys_properties] table
delta_page_size	Enter the page size, that is, the maximum number of records fetched per page in a data pull for the data sources within the connector. - Type: integer - Default value: 500 - Location: System Property [sys_properties] table
insert_short_lived_vms	Set the property to <code>false</code> to filter out Azure Databricks Cls during delta loads. For more information, see Azure Databricks Cls. - Type: true false - Default value: true - Location: System Property [sys_properties] table
is_delta_sync_enabled	Set the property to <code>false</code> to deactivate delta synchronization. - Type: true false - Default value: false - Location: System Property [sys_properties] table

Property	Description
	See Configuring data synchronization in Service Graph Connector for Microsoft Azure .
LDC_status	Set the property to <code>true</code> to enable populating the passive status of an Azure datacenter (also known as a logical datacenter) in the Comments attribute of the Azure Datacenter [cmdb_ci_azure_datacenter] CI class when the <code>regionType</code> attribute is <code>Logical</code> or the service account for the datacenter is deactivated . • Type: <code>true</code> <code>false</code> • Default value: <code>false</code> • Location: System Property [sys_properties] table
load_balancer_page_limit	Enter the maximum number of records per page to be fetched in a delta pull for the SG-Azure Load Balancers data source. • Type: <code>integer</code> • Default value: <code>500</code> • Location: System Property [sys_properties] table
maximum_delta_total_period	Enter the maximum time period in days preceding the current run time during delta pull.

Property	Description
	- Type: integer - Default value: 7 - Location: System Property [sys_properties] table
maximum_delta_window_period	Enter the maximum time duration in hours for each API call to repeat within the total delta period during delta pull. - Type: integer - Default value: 24 - Location: System Property [sys_properties] table
network_interface_page_limit	Enter the maximum number of records per page to be fetched in a delta pull for the SG-Azure Network Interface data source. - Type: integer - Default value: 500 - Location: System Property [sys_properties] table
network_page_limit	Custom property for the SG-Azure Network data source. Create this system property if you need to decrease the number of records that are fetched in a delta pull. For more information, see Add a system property.

Property	Description
	Specify the maximum number of records per page to be fetched in a delta pull for the SG-Azure Network data source. • Type: integer • Default value: 500 • Location: System Property [sys_properties] table
pull_disabled_subscriptions	Set the property to <code>false</code> to deactivate the import of resources associated with deactivated Azure subscriptions. • Type: <code>true</code> <code>false</code> • Default value: <code>true</code> • Location: System Property [sys_properties] table
security_group_page_limit	Enter the maximum number of records per page to be fetched in a delta pull for the SG-Azure Security Group data source. • Type: integer • Default value: 500 • Location: System Property [sys_properties] table
software_page_limit	Enter the maximum number of records fetched per page when importing software data through

Property	Description
	the SG-Azure Software data source. - Type: integer - Default value: 5000 - Location: System Property [sys_properties] table
tcp_and_process_page_limit	Enter the maximum number of records fetched per page when importing TCP and running process data through the SG-Azure TCP data source. - Type: integer - Default value: 1000 - Location: System Property [sys_properties] table
tcp_time_period	Enter the time period in hours during which TCP and running process data is imported through the SG-Azure TCP data source. - Type: integer - Default value: 1 - Location: System Property [sys_properties] table
vm_power_state_page_limit	Enter the maximum number of records fetched per page when importing the power state of VMs through the SG-Azure VM Config Data data source.

Property	Description
	- Type: integer - Default value: 1000 - Location: System Property [sys_properties] table

Connection property

The filter_databricks connection property is available for the Hardware connections of the Service Graph Connector for Microsoft Azure.

Note: To open the Service Graph Connection Properties [sn_cmdb_int_util_service_graph_connection_property] table for the connector, navigate to **All > Service Graph Connectors > Azure > Connections**, and select the connection name. The connection properties are displayed in the Service Graph Connection Properties related list.

filter_databricks

Set the value to `true` for each Hardware connection to filter Azure Databricks CIs. For more information, see [Azure Databricks CIs](#).

- Type: true | false
- Default value: `false`

Azure Databricks CIs

Azure Databricks virtual machines (VMs), Network Interface Controllers (NICs), and storage volumes (collectively called Azure Databricks CIs) have the value of the Vendor tag set to `Databricks`. Starting with the Service Graph Connector for Microsoft Azure 1.14.0 version, all Microsoft Azure Databricks CIs that have the value of the Vendor tag set to `Databricks` are filtered out from discovery during a full data load.

During delta loads, set the value of the insert_short_lived_vms system property to `false` to filter Azure Databricks CIs that are marked as retired during discovery. Additionally, for each Hardware connection, set the filter_databricks connection property to `true`.

